

Hyperkalemia — treatment (stabilize, shift, remove)

1 CALCIUM

2 INSULIN/ GLUCOSE

3 ALBUTEROL

4 ± BICARB (if NAGMA)

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5 DIALYSIS/ DIURETIC/ BINDER

21 EMERGENCY ORDER:
Calcium → Insulin/Glucose
→ Albuterol →
± Bicarb → Removal

THE BANANA EMERGENCY PHARMACY — TREATMENT OF HYPERKALEMIA

STABILIZE → SHIFT → REMOVE — three tiers, three jobs

1 TIER 1: STABILIZE — when ECG changes or $K^+ \geq 6.5$

2

4 Calcium gluconate
10 mL of 10% IV over 2-3 min — peripheral OK

4 $CaCl_2$
central line only with a "CENTRAL LINE" warning

5 1-3 min
30-60 min

6 DIGOXIN WARNING
DIGOXIN TOXICITY: give calcium SLOWLY, diluted, over 20-30 min

7 TIER 2: SHIFT — temporary, body K^+ unchanged

8

9 NEVER GLUCOSE ALONE

10 Monitor glucose — D10 Arip if needed

11 10-20 mg opb
 β_2
ALBUTEROL 10-20 mg neb — additive, never alone

12 ESRD may resist

12 ONLY IF NAGMA
 $NaHCO_3$
AVOID BOLUS

12 ESRD may be RESISTANT to β_2 — combine with insulin

14 TIER 3: REMOVE — only this lowers TOTAL body K^+

15

16 PD much less effective

17 LOOP DIURETIC — needs functioning kidney; saline + furosemide + distal Na^+

18 SPS/KAYEXALATE
— slow, risks of necrosis (axe, past-up, flous)

19 PATIROMER
— chronic use; side hypomagnesemia

20 SDC
sodium zirconium cyclosilicate (SEE)
fastest, safest binder, acute + chronic

THE PHARMACY PEARL SCROLL

STABILIZE WITH CALCIUM. SHIFT WITH INSULIN/GLUCOSE + ALBUTEROL. REMOVE WITH DIALYSIS / DIURETIC / BINDER.

SHIFT IS TEMPORARY — REMOVAL IS DEFINITIVE.

THE TRIAGE LADDER

THE STABLE OUTPATIENT CART

6 DIGOXIN WARNING
STABLE: offending drug

22 STABLE: stop offending drugs (ACEI/ARB/spironolactone/NSAID/TMP)

23 Low- K^+ diet — AVOID salt substitutes (KCl)

24 Chronic binder + follow-up labs

THE STABLE OUTPATIENT CART

1. 3 tiers

WHAT YOU SEE

Banner: stabilize + shift + remove

WHAT IT ENCODES

Hyperkalemia management has three sequential jobs: (1) STABILIZE the myocardial membrane (calcium), (2) SHIFT K⁺ into cells temporarily (insulin/glucose, albuterol, +/- bicarb), and (3) REMOVE K⁺ from the body (diuretics, binders, dialysis). Only the third job lowers total body potassium; the first two buy time. Severity (ECG changes, K⁺ level) dictates how aggressively you move through the tiers.

BOARD TRAP

WRONG answer: jumping straight to insulin or kayexalate when the ECG shows peaked T waves. Discriminator: with ECG changes you must give CALCIUM first to protect the heart before anything else — shifting and removing K⁺ take minutes to hours and won't prevent imminent arrhythmia.

2. Tier 1: stabilize

WHAT YOU SEE

Tier 1 shelf, alarm light

WHAT IT ENCODES

Tier 1 (membrane stabilization) is indicated when there are ECG changes (peaked T waves, then PR prolongation/widened QRS, loss of P waves, sine wave) OR when K⁺ is severe (commonly K⁺ >= 6.5). Calcium raises the threshold potential, restoring the gap between resting and threshold membrane potential that hyperkalemia had narrowed.

BOARD TRAP

WRONG answer: waiting for a 'high enough' number before treating a patient with ECG changes. Discriminator: ANY ECG manifestation of hyperkalemia is an emergency warranting immediate calcium regardless of the exact K⁺ value — the rhythm, not just the number, drives urgency.

3. IV calcium gluconate

WHAT YOU SEE

Calcium gluconate vials, hero

WHAT IT ENCODES

IV calcium gluconate (typically 10 mL of 10% over 2–3 min, repeat in 5 min if ECG unchanged) is the FIRST drug given for hyperkalemia with ECG changes. It antagonizes the membrane effects of hyperkalemia within 1–3 minutes; the effect lasts only ~30–60 minutes, so it must be paired with shifting and removal.

BOARD TRAP

WRONG answer: believing calcium lowers the serum potassium. Discriminator: calcium does NOT change the K⁺ level at all — it only stabilizes the myocardium. A learner who gives calcium and stops has failed to treat the hyperkalemia; you still must shift and remove.

4. Central line for CaCl

WHAT YOU SEE

Central line note for calcium chloride

WHAT IT ENCODES

Calcium chloride contains about 3x more elemental calcium than calcium gluconate (~270 mg vs ~90 mg per 10 mL ampule), so it is preferred in cardiac arrest/peri-arrest, but it is highly caustic and should be given through a CENTRAL line to avoid severe tissue necrosis on extravasation. Calcium gluconate is safer peripherally for routine use.

BOARD TRAP

WRONG answer: pushing calcium chloride through a small peripheral IV in a stable ward patient. Discriminator: extravasation causes severe local necrosis — use calcium gluconate peripherally; reserve CaCl for central access or a true arrest situation.

5. Digoxin caution

WHAT YOU SEE

Digoxin toxicity warning on calcium

WHAT IT ENCODES

When hyperkalemia coexists with digoxin toxicity, give calcium cautiously and SLOWLY (over 20–30 min) because of the theoretical 'stone heart' — calcium added to digoxin-induced intracellular calcium overload may precipitate refractory contraction/arrhythmia. The definitive treatment is digoxin-specific antibody (Fab) fragments, which also correct the hyperkalemia.

BOARD TRAP

WRONG answer: rapidly bolusing IV calcium in a digoxin-toxic patient with hyperkalemia. Discriminator: classic teaching is to avoid/give calcium very cautiously here and prioritize digoxin-immune Fab — the hyperkalemia reflects pump poisoning that Fab reverses at its source.

6. Tier 2: shift

WHAT YOU SEE

Tier 2 shelf — shift, temporary

WHAT IT ENCODES

Tier 2 drives K⁺ from the extracellular space into cells via the Na/K-ATPase. It works in minutes and lasts a few hours, but total body K⁺ is UNCHANGED — K⁺ will leak back out as the agents wear off. Shifting is a bridge to definitive removal, not a cure, especially in anuric/ESRD patients who cannot excrete the load.

BOARD TRAP

WRONG answer: declaring the patient treated after the K⁺ drops with insulin/albuterol. Discriminator: shifting is temporary and total body K⁺ is the same — in a patient who cannot excrete K⁺ (oliguric AKI, ESRD), you MUST proceed to removal (dialysis) or the K⁺ will rebound.

7. Insulin + glucose

WHAT YOU SEE

Insulin 10 U + 25 g glucose vials

WHAT IT ENCODES

Regular insulin 10 units IV with 25 g of dextrose (one ampule of D50) is the fastest, most reliable shifting therapy; insulin activates the Na/K-ATPase to drive K⁺ into cells, lowering serum K⁺ by ~0.5–1.2 mEq/L within 15–30 minutes for several hours. Glucose is co-administered to prevent iatrogenic hypoglycemia.

BOARD TRAP

WRONG answer: omitting the glucose 'because the patient got insulin.' Discriminator: hypoglycemia is a common, sometimes severe complication 30–60 min later — always give dextrose (and monitor glucose), EXCEPT when the patient is already significantly hyperglycemic (e.g., glucose > 250), in which case give insulin alone and recheck.

8. Albuterol

WHAT YOU SEE

Albuterol inhaler on ladder

WHAT IT ENCODES

High-dose nebulized albuterol (a beta-2 agonist, typically 10–20 mg — about 4–8x the bronchodilator dose) stimulates the Na/K-ATPase and shifts K⁺ into cells, lowering serum K⁺ by ~0.5–1.0 mEq/L. Its effect is ADDITIVE to insulin, making the two a standard combined shifting regimen.

BOARD TRAP

WRONG answer: relying on albuterol as monotherapy. Discriminator: up to ~40% of patients (especially those on beta-blockers) are poor responders, and the doses needed cause tachycardia — use it as an adjunct to insulin/glucose, not a substitute.

9. Bicarbonate

WHAT YOU SEE

NaHCO₃ — only if NAGMA

WHAT IT ENCODES

Sodium bicarbonate raises pH, which can shift K⁺ into cells in exchange for H⁺, but it is slow and weak and only meaningfully helps when a metabolic acidosis is present. It is most useful as an infusion in acidotic CKD patients rather than as an acute bolus.

BOARD TRAP

WRONG answer: giving a bicarbonate bolus as first-line for isolated, non-acidotic hyperkalemia. Discriminator: bicarbonate is NOT first-line and is largely ineffective without acidosis; in ESRD it risks volume overload and hypocalcemia/tetany and does not reliably lower K⁺ acutely.

10. Tier 3: remove

WHAT YOU SEE

Tier 3 shelf — remove, lowers total K⁺

WHAT IT ENCODES

Tier 3 is the only category that lowers TOTAL BODY potassium: enhance renal excretion (loop diuretic), pull K⁺ out via the gut (binders), or extract it directly (dialysis). Every hyperkalemia patient ultimately needs a removal strategy once the membrane is stabilized and K⁺ is temporarily shifted.

BOARD TRAP

WRONG answer: treating with stabilize + shift only and discharging without a removal plan. Discriminator: without removal, shifted K⁺ returns within hours — match the removal modality to the cause (diuretic if kidneys work, dialysis if they don't, binders for chronic control).

11. Dialysis

WHAT YOU SEE

Dialysis apothecary, most effective

WHAT IT ENCODES

Hemodialysis is the most effective and definitive removal method and is the treatment of choice for severe or refractory hyperkalemia, ESRD, or oliguric AKI. It can remove large amounts of K⁺ rapidly; note that K⁺ can rebound after a session as it redistributes from cells, so recheck levels.

BOARD TRAP

WRONG answer: delaying dialysis in an anuric ESRD patient while cycling more insulin and albuterol. Discriminator: if the kidneys cannot excrete K⁺, shifting only buys time — dialysis is the definitive answer and should be arranged early in severe/refractory cases.

12. Loop diuretic

WHAT YOU SEE

Loop diuretic on remove shelf

WHAT IT ENCODES

Loop diuretics (e.g., furosemide) increase distal Na⁺ delivery and urine flow, enhancing renal K⁺ excretion — but they REQUIRE functioning kidneys and adequate volume status to work. They are useful in volume-overloaded patients with preserved renal function and as adjuncts in chronic management.

BOARD TRAP

WRONG answer: ordering furosemide to fix hyperkalemia in an anuric/ESRD patient. Discriminator: no urine output means no diuretic-mediated K⁺ loss — in advanced renal failure the loop diuretic is ineffective and dialysis is required instead.

13. K⁺ binders

WHAT YOU SEE

SPS/kayexalate, patiomer, SZC bottles

WHAT IT ENCODES

GI cation exchangers bind K⁺ in the gut and remove it in stool, lowering total body K⁺: patiomer and sodium zirconium cyclosilicate (SZC) are newer agents, while sodium polystyrene sulfonate (SPS/kayexalate) is the older one. SZC has the fastest onset of the binders (~1 hour) and is useful for both acute and chronic control; patiomer and SPS are slower (better for subacute/chronic).

BOARD TRAP

WRONG answer: relying on kayexalate for EMERGENCY hyperkalemia. Discriminator: SPS works slowly (hours) and carries a risk of colonic/intestinal NECROSIS (especially with sorbitol, post-op, or bowel disease) — it is not a substitute for calcium/insulin in a crisis; patiomer and SZC are newer and generally safer for ongoing management.

14. Stop offending drugs

WHAT YOU SEE

Stable outpatient cart, stop drugs

WHAT IT ENCODES

For the stable outpatient, the foundation of management is removing contributors: stop or reduce ACEi/ARBs, potassium-sparing diuretics (spironolactone/eplerenone/amiloride), NSAIDs, and trimethoprim; restrict dietary potassium; and counsel patients to AVOID salt substitutes (which are potassium chloride). Chronic binders (patiomer/SZC) can allow continuation of needed RAAS therapy in heart failure/CKD.

BOARD TRAP

WRONG answer: telling a hyperkalemic patient to use a 'low-sodium' salt substitute for blood pressure. Discriminator: salt substitutes are KCl and are a hidden, common cause of outpatient hyperkalemia — review the med and diet list, including OTC supplements, before escalating therapy.